

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Theory of Computation (New)

Semester: Spring

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain Mathematical Induction. Using mathematical induction prove that sum of first n positive odd numbers $1+3+5+\dots+2n-1 = n^2$. 7
- b) Differentiate Between DFA and NFA? Design a DFA that accepts the language given by $L=\{w \in \{0,1\}^* : w \text{ contains '00' or '11' as substring}\}$. Hence test your design for 101001 and 0101010. 8
2. a) Minimize the following DFA (Draw initial diagram first). Specify performed operations in each step. 7

δ/Σ	0	1
$\rightarrow Q0$	Q1	Q2
*Q1	Q1	Q3
*Q2	Q2	Q2
*Q3	Q5	Q2
*Q4	Q4	Q2
*Q5	Q4	Q2
Q6	Q5	Q6
Q7	Q5	Q6
Note: $\rightarrow$ for start state, * for final state		

- b) State Pumping Lemma for regular set. Use pumping lemma to prove that the language $L=\{0^n1^n : n>0\}$. 8
3. a) What is GNF? Convert the following CFG into GNF. 7
 $S \rightarrow XA|BB$
 $B \rightarrow b|SB$
 $X \rightarrow b$
 $A \rightarrow a$

- b) Design a PDA which accepts the language $L = \{w \in \{0,1\}^* : w \text{ has equal number of 0's and 1's}\}$ and also test your design for strings "**010110**" and "**11010**". 8

OR

Design a PDA which accepts the language $L = \{a^i b^j c^k : k = i+j\}$ and also test your design for strings "**aabcccc**" and "**bbcccc**".

4. a) When is the grammar is said to be ambiguous? Prove the grammar is ambiguous: $S \rightarrow 0S1 \mid 1S0 \mid SS \mid \epsilon$. 7
- b) What is Unrestricted Grammar? Design a Turing Machine to accept the language $L = \{0^n 1^n : n \geq 1\}$. 8

OR

Formally define a Turing Machine $M = (Q, \Sigma, \Gamma, \delta, q_0, B, F)$ that accepts the language: $L = \{w \in \{a, b\}^* \mid w \text{ is a palindrome}\}$. Trace the steps of the machine when processing the input string abba.abba.

5. a) How does Turing Machine compute a function? Design a Turing machine to compute a function $f(m) = m+1$ where m belongs to set of Natural numbers. 7
- b) What is recursive and recursively enumerable language? Show that the intersection of two recursive language is also recursive. 8
6. a) Explain the Church-Turing thesis. How does the Post correspondence problem (PCP) demonstrate computational undecidability? 7
- b) What is meant by time and space complexity? Explain the classes P and NP in detail with examples. How are they related to real-world problem-solving? 8
7. Write short notes on: (**Any two**) 2×5
- a) Pigeonhole principle
- b) Pumping Lemma for CFL
- c) Universal Turing Machine